

Day-ahead PV Forecasting with a Novel Physics-Constrained Transformer Model Design

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Abstract—Day-ahead photovoltaic (PV) forecasting is essential for reliable grid scheduling and renewable energy integration, yet it remains difficult because PV generation is highly intermittent, nonlinear, and sensitive to changing weather conditions. Existing methods often rely on future meteorological inputs or lack physical feasibility. To address this, this paper proposes a physics-constrained Transformer in a history-only setting, integrating weather-regime-aware clustering, cross-channel dependency modeling, and feasibility-oriented constraints. Comparative experiments demonstrate effective and robust next-day PV power forecasting performance.

Index Terms—Photovoltaic power forecasting, Optimization, Renewable Energy Integration, Uncertainty modeling, Smart Grids

I. INTRODUCTION

This introduction should be rewritten

The intermittency and uncertainty of photovoltaic (PV) power generation pose significant challenges to grid stability, scheduling, and day-ahead dispatch operations. Therefore, improving the accuracy of day-ahead PV power forecasting is essential for enhancing dispatch quality and maintaining reliable system operation. In practical deployment, however, continuous access to high-quality numerical weather prediction (NWP) inputs may not always be guaranteed, especially in data-constrained or resource-limited settings. Motivated by such scenarios, this work studies day-ahead PV forecasting under a history-only setting, where predictions are generated solely from historical observations available up to the forecast origin.

Existing PV forecasting approaches mainly include physical models, statistical methods, and hybrid learning frameworks. Physical models can provide coarse-grained descriptions of atmospheric and electrical processes, but they often struggle to produce accurate forecasts when PV generation exhibits strong nonlinearity, intermittency, and variability [1]. Statistical methods, including time-series analysis [2] and regression-based approaches [3], [4], are often limited in their ability to

capture complex nonlinear relationships and may accumulate substantial errors over long forecasting horizons. To address these limitations, hybrid frameworks have combined deep learning [5], signal decomposition, and optimization strategies into integrated pipelines for PV forecasting. For example, Liu *et al.* [6] employed a CNN-GRU model to extract spatiotemporal features from multidimensional time series, while Li *et al.* [7] introduced temporal convolutional networks (TCN) to enlarge the receptive field and further incorporated attention mechanisms to improve predictive accuracy [8]. Although such hybrid models can improve performance, simply stacking multiple components does not necessarily provide a coherent framework that fully exploits their complementary strengths [9]. Similarly, bidirectional LSTM based models can enhance contextual feature extraction within observed sequences [10], but their forecasting performance may still be limited when long-range dependencies and multivariate interactions become increasingly complex.

Recently, Transformer-based architectures [11] have emerged as a dominant paradigm in time-series forecasting because self-attention mechanisms can model long-range dependency patterns effectively. To improve efficiency and adapt to multivariate time-series (MTS) forecasting, several variants such as Informer [12] and Autoformer [13] have been proposed. In solar forecasting, Piantadosi *et al.* [15] developed a Transformer-based framework that incorporates plant configuration and air-mass related information, achieving lower PV power prediction errors than traditional recurrent models. Zhang *et al.* [16] further proposed a Mask-Transformer to improve robustness under missing meteorological observations. Despite these advances, many existing Transformer-based forecasters primarily emphasize temporal dependency modeling, while the cross-variable interactions among irradiance components, meteorological conditions, temporal context, and plant dynamics are not always modeled explicitly or efficiently [14]. This limitation

is especially important for PV forecasting, where the coupling among multiple physical and environmental variables strongly influences generation behavior.

Meanwhile, purely data-driven deep learning models often lack explicit feasibility control and weakly reflect physical operating characteristics [17]. Because solar irradiance is highly sensitive to multiscale cloud variability, observation-based statistical models often degrade under complex cloudy conditions. This motivates the incorporation of physics-constrained elements into the forecasting framework. For example, the clear-sky index has been widely used to normalize irradiance and remove the deterministic diurnal solar cycle [19]. To better characterize cloud effects, Xie and Liu [20] discussed the extraction of cloud fraction and cloud albedo from shortwave radiation measurements, while Liu *et al.* [18] proposed a persistence-based forecasting framework that uses Relative Cloud Radiative Forcing (RCRF) to jointly model GHI, DNI, and DHI. These studies suggest that physically grounded radiation descriptors can provide useful structural guidance for PV forecasting. However, existing methods still lack a unified framework that combines multivariate cross-channel learning, weather-regime-aware training, and feasibility-oriented physical constraints in a history-only day-ahead forecasting setting.

To address these issues, this paper proposes a physics-constrained forecasting framework for day-ahead PV power prediction under a history-only setting. The method first constructs multivariate input features from historical irradiance, meteorological variables, temporal encodings, and physics-constrained reference features. A training-time weather pattern clustering module is then introduced to identify recurrent cloud–radiation regimes and improve regime coverage through cluster-balanced mini-batch sampling. Based on these representations, an improved lightweight Transformer architecture is designed to model cross-channel dependencies among observed variables, while a Fourier-inspired nonlinear transformation is selectively applied to enhance the representation of smooth and quasi-periodic generation patterns. Finally, feasibility-aware penalties and post-processing projection are incorporated to enforce non-negativity, capacity consistency, and historically calibrated ramp behavior. In this way, the proposed framework aims to improve robustness and structural consistency for day-ahead PV forecasting in settings where future meteorological forecast inputs are unavailable.

II. GENERAL SCHEME OF THE MODEL

A. Data

The high-resolution photovoltaic (PV) operational data utilized in this study are sourced from the Yulara site, administered by the Desert Knowledge Australia (DKA) Solar Centre. Specifically, this research employs data from the roof-mounted distributed PV system designated as “Site 3-A” (Sails in the Desert - 1). Commissioned in 2016, the system features a rated installed capacity of 22.6 kW and utilizes monocrystalline silicon (mono-Si) module technology. The resolution of the dataset is 10 minutes, 144 steps per day. This open-access dataset provides high-quality, real-world time-series records of

actual PV power generation. Integrated with localized meteorological conditions, it serves as a robust empirical foundation for training and validating the proposed framework for day-ahead solar forecasting [21].

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B. Feature Construction and Physics-Constrained Priors

To study day-ahead PV forecasting under a history-only setting, the input representation is constructed solely from historical observations available up to the forecast origin, without using numerical weather prediction (NWP) inputs or any future meteorological information. This setting is also used to examine how much predictive information can be extracted from historical observations alone in data-constrained forecasting scenarios. Among the meteorological variables, global horizontal irradiance (GHI), direct normal irradiance (DNI), and diffuse horizontal irradiance (DHI) are considered jointly because they describe complementary radiative components governing PV generation [18]. Specifically, GHI denotes the total shortwave irradiance incident on the horizontal plane, DNI denotes the direct beam component measured on the normal plane, and DHI denotes the diffuse component mainly associated with atmospheric scattering and cloud effects. Using all three components provides a more physically grounded description of the radiation field driving PV power output. When DHI is not directly measured, it is reconstructed using the irradiance balance relation

$$DHI_t = \max(0, GHI_t - DNI_t \max(0, \cos \theta_t^z)), \quad (1)$$

where θ_t^z is the solar zenith angle at time step t . This formulation is consistent with the standard projection from direct normal irradiance to the horizontal plane.

Based on these considerations, the multivariate input vector at time step t is defined as

$$\mathbf{x}_t = [GHI_t, DNI_t, DHI_t, T_t, p_t, s_t^h, c_t^h, s_t^d, c_t^d, P_{ic,t}, \Delta P_t, P_t] \in \mathbb{R}^D, \quad (2)$$

with $D = 12$. Here, T_t and p_t denote air temperature and atmospheric pressure, respectively; $s_t^h = \sin(2\pi h_t/24)$ and $c_t^h = \cos(2\pi h_t/24)$ are the sine and cosine encodings of the hour-of-day index h_t ; $s_t^d = \sin(2\pi d_t/T_{\text{year}})$ and $c_t^d = \cos(2\pi d_t/T_{\text{year}})$ are the sine and cosine encodings of the day-of-year index d_t , where T_{year} denotes the annual period used in the calendar encoding; P_t denotes the observed historical PV power output; and $\Delta P_t = P_t - P_{t-1}$ denotes the short-term power ramp.

To further incorporate a compact physics-informed reference while avoiding information leakage, an irradiance-curve baseline is introduced as

$$P_{ic,t} = f_{ic}(GHI_t), \quad (3)$$

where $f_{ic}(\cdot)$ approximates the plant’s typical power response under observed irradiance conditions. The mapping $f_{ic}(\cdot)$ is estimated only from the training split using a binned-median fitting strategy, and $P_{ic,t}$ is computed only for timestamps within the historical input window. Therefore, $P_{ic,t}$ serves

as a history-dependent reference feature rather than a future covariate. Finally, all feature channels are standardized using z-score statistics computed exclusively from the training data.

C. Phase 1: Physics-Constrained Weather Pattern Clustering

To improve training robustness under diverse meteorological conditions, we introduce a physics-constrained deep clustering module as a training-time regime discovery component. The purpose of this module is to identify recurrent weather regimes from historical daily radiation patterns and to use the resulting regime labels to rebalance the training process of the forecasting model. Because the future weather regime is unknown at prediction time, the clustering outputs are used only during training and are never provided to the forecasting network as inference-time inputs.

For each historical day, we construct a daily physical descriptor sequence using radiation-related variables that characterize cloud–radiation behavior. In particular, we use the clear-sky index and the relative cloud radiative forcing, defined as

$$K_{cs,t} = \frac{\text{GHI}_t}{\text{GHI}_t^{cs} + \varepsilon}, \quad \text{RCRF}_t = \frac{\text{GHI}_t^{cs} - \text{GHI}_t}{\text{GHI}_t^{cs} + \varepsilon}, \quad (4)$$

where GHI_t^{cs} denotes the clear-sky global horizontal irradiance at time step t , and $\varepsilon > 0$ is a small constant introduced for numerical stability. Let

$$\mathbf{W} = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_N\} \quad (5)$$

denote the collection of daily descriptor sequences, where $\mathbf{w}_i \in \mathbb{R}^{T_d \times D_w}$ is the descriptor sequence of the i -th historical day, T_d is the number of time steps within one day, D_w is the descriptor dimension, and N is the number of training days.

An encoder f_θ maps each daily descriptor sequence into a latent representation,

$$\mathbf{z}_i = f_\theta(\mathbf{w}_i) \in \mathbb{R}^{d_z}, \quad (6)$$

where $\mathbf{z}_i$ is a fixed-dimensional latent vector summarizing the cloud–radiation dynamics of the i -th day. In our implementation, f_θ first processes the input sequence along the temporal dimension and then aggregates the hidden sequence into a single vector representation through temporal pooling.

To jointly optimize latent representation learning and regime discovery, clustering is performed in the latent space using a Kullback–Leibler (KL) divergence based objective. The soft assignment probability between latent representation $\mathbf{z}_i$ and the j -th cluster center μ_j is defined by the Student's t -distribution:

$$q_{ij} = \frac{(1 + \|\mathbf{z}_i - \mu_j\|^2/\alpha)^{-\frac{\alpha+1}{2}}}{\sum_{c=1}^C (1 + \|\mathbf{z}_i - \mu_c\|^2/\alpha)^{-\frac{\alpha+1}{2}}}, \quad (7)$$

where C is the number of clusters and α is the degrees-of-freedom parameter. Based on q_{ij} , the auxiliary target distribution is constructed as

$$p_{ij} = \frac{q_{ij}^2/f_j}{\sum_{c=1}^C q_{ic}^2/f_c}, \quad f_j = \sum_{i=1}^N q_{ij}, \quad (8)$$

which encourages more compact and discriminative regime partitions. The phase-1 objective is defined as

$$\mathcal{L}_{\text{phase1}} = \mathcal{L}_{\text{rec}} + \gamma \text{KL}(\mathcal{P} \parallel \mathcal{Q}) + \lambda \mathcal{L}_{\text{bal}}, \quad (9)$$

where $\mathcal{L}_{\text{rec}}$ is the sequence reconstruction loss, $\mathcal{L}_{\text{bal}}$ is a cluster-balance regularizer, and $\gamma, \lambda > 0$ are weighting coefficients.

The resulting cluster assignments characterize historical weather regimes with similar cloud–radiation structures. These regime labels are then used only during forecasting-model training to perform cluster-balanced mini-batch sampling, so that each mini-batch contains a more uniform exposure to different meteorological regimes rather than being dominated by the most frequent weather pattern. In this way, the clustering module improves regime coverage and training robustness under heterogeneous meteorological conditions, while preserving a strictly history-only forecasting setup. At test time, no regime label or latent clustering representation is provided to the forecasting model.

D. Phase 2: Proposed Physics-Constrained Modified Transformer Architecture

Let $\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L \times D}$ denote the historical input window of length L , where D is the number of feature channels and each row corresponds to the multivariate observation vector at one historical time step. The forecasting objective is to predict the next-day PV power trajectory

$$\hat{\mathbf{y}}_{t+1:t+H} = [\hat{P}_{t+1}, \hat{P}_{t+2}, \dots, \hat{P}_{t+H}]^\top \in \mathbb{R}^H, \quad (10)$$

using only observations available up to the forecast origin t . To this end, we employ an improved lightweight Transformer architecture for multivariate PV forecasting under a strictly history-only setting. Instead of relying on heavy temporal self-attention over the entire sequence, the model emphasizes cross-channel interaction, which is more directly aligned with the coupling among irradiance components, meteorological variables, temporal context, and plant power dynamics.

Specifically, the target power sequence is separated from the remaining observed features. Let

$$\mathbf{P}_{t-L+1:t} \in \mathbb{R}^{L \times 1} \quad (11)$$

denote the historical PV power sequence, and let

$$\mathbf{X}_{t-L+1:t}^{\text{obs}} \in \mathbb{R}^{L \times (D-1)} \quad (12)$$

denote the matrix of remaining non-target observed features, including irradiance-related variables, meteorological variables, temporal encodings, and physics-constrained auxiliary features. These two parts are projected into latent embeddings as

$$\mathbf{E}^{\text{tar}} = \mathbf{P}_{t-L+1:t} \mathbf{W}_{\text{tar}}, \quad \mathbf{E}^{\text{obs}} = \mathbf{X}_{t-L+1:t}^{\text{obs}} \mathbf{W}_{\text{obs}}, \quad (13)$$

where $\mathbf{W}_{\text{tar}} \in \mathbb{R}^{1 \times d_{\text{model}}}$ and $\mathbf{W}_{\text{obs}} \in \mathbb{R}^{(D-1) \times d_{\text{model}}}$ are learnable projection matrices, and therefore $\mathbf{E}^{\text{tar}}, \mathbf{E}^{\text{obs}} \in \mathbb{R}^{L \times d_{\text{model}}}$. The two embeddings are then concatenated along the token/channel dimension to form

$$\mathbf{E} = [\mathbf{E}^{\text{tar}}; \mathbf{E}^{\text{obs}}], \quad (14)$$

which is subsequently processed by a stack of M lightweight sequence blocks:

$$\mathbf{H} = \frac{1}{M} \sum_{m=1}^M \mathcal{B}_m(\mathbf{E}), \quad (15)$$

where $\mathcal{B}_m(\cdot)$ denotes the m -th sequence block and $\mathbf{H}$ is the fused hidden representation.

Within each block, channel-wise interaction is used to model dependencies among radiation variables, thermodynamic conditions, temporal context, and historical power signals. In addition, a Fourier Activation Network (FAN) style feature transformation module is selectively applied to continuous latent feature representations in order to enhance the modeling of smooth and quasi-periodic variations commonly observed in solar generation data. For an intermediate latent feature vector $\mathbf{u} \in \mathbb{R}^{d_{model}}$, the FAN-style transformation is written in the generic form

$$\phi_{\text{FAN}}(\mathbf{u}) = W_r[\cos(W_f \mathbf{u}), \sin(W_f \mathbf{u}), \sigma(W_g \mathbf{u})], \quad (16)$$

where $W_f \in \mathbb{R}^{d_f \times d_{model}}$, $W_g \in \mathbb{R}^{d_g \times d_{model}}$, $W_r \in \mathbb{R}^{d_{model} \times (2d_f + d_g)}$ are learnable projection matrices, and $\sigma(\cdot)$ is a nonlinear activation function. This transformation is not applied indiscriminately to all raw input channels, since some temporal variables are already represented through explicit sine-cosine encodings and therefore already preserve their periodic structure. Instead, it is applied only after linear embedding, on selected continuous latent representations within each sequence block. To preserve dimensional consistency for residual connections, the concatenated FAN output is linearly projected back to the original latent dimension.

Finally, the fused representation is passed to a temporal decoder to generate the normalized next-day prediction,

$$\hat{\mathbf{y}}_{t+1:t+H} = W_o \mathcal{D}(\mathbf{H}), \quad (17)$$

where $\mathcal{D}(\cdot)$ denotes the decoder and W_o is the output projection matrix. In this way, the forecasting backbone remains fully history-only at inference time, while still capturing multivariate cross-channel dependencies and periodic structure in a compact architecture.

E. Robust Optimization and Physics-Constrained Inference

Time-series forecasting models with large parameter spaces may overfit noisy fluctuations and produce physically implausible outputs, such as negative generation or values exceeding the plant capacity. To improve optimization robustness while preserving physical feasibility, the loss is computed on denormalized power values in the original physical unit. In addition, Sharpness-Aware Minimization (SAM) is adopted to improve flatness and generalization under noisy training dynamics, while physical plausibility is primarily enforced by denormalized training objectives and feasibility-aware penalties and post-processing projection.

Let $\hat{P}_{t+h}$ denote the denormalized prediction and P_{t+h} denote the corresponding ground-truth PV power at horizon step $h \in \{1, \dots, H\}$. Let P_r denote the rated capacity of the

PV plant. In this study, the rated capacity is known *a priori* and fixed at $P_r = 22.6$ kW. The primary forecasting loss is defined as the capacity-normalized mean squared error

$$\mathcal{L}_{\text{power}} = \frac{1}{P_r^2 H} \sum_{h=1}^H \left(\hat{P}_{t+h} - P_{t+h} \right)^2. \quad (18)$$

To discourage infeasible predictions, two soft constraint penalties are introduced. The first is the capacity penalty

$$\mathcal{L}_{\text{cap}} = \frac{1}{P_r^2 H} \sum_{h=1}^H \max\left(0, \hat{P}_{t+h} - P_r\right)^2, \quad (19)$$

which penalizes predictions that exceed the plant rating. The second is the non-negativity penalty

$$\mathcal{L}_{\text{nonneg}} = \frac{1}{P_r^2 H} \sum_{h=1}^H \max\left(0, -\hat{P}_{t+h}\right)^2, \quad (20)$$

which penalizes negative generation values. The overall training objective is then

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{power}}(\theta) + \eta (\mathcal{L}_{\text{cap}}(\theta) + \mathcal{L}_{\text{nonneg}}(\theta)), \quad (21)$$

where θ denotes the model parameters and $\eta = 0.1$ is an empirically chosen penalty weight that provides moderate regularization without dominating the primary forecasting loss.

To improve training stability, we optimize this objective using SAM, which seeks model parameters located in flatter neighborhoods of the loss landscape:

$$\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}_{\text{total}}(\theta + \epsilon), \quad (22)$$

where ϵ is an adversarial parameter perturbation and ρ is the perturbation radius. In practice, this min-max update is approximated with the standard first-order SAM procedure, which requires two forward-backward evaluations per optimization step. Here, SAM is used as a robustness-oriented optimization strategy to reduce sensitivity to sharp local minima and improve generalization, rather than as a dedicated physical constraint mechanism.

During inference, an additional feasibility-aware post-processing step is applied. The raw prediction is first clipped into the admissible generation interval

$$\bar{P}_{t+h} = \min\left(P_r, \max(0, \hat{P}_{t+h})\right). \quad (23)$$

To further limit implausible step-to-step fluctuations, a ramp reference R is estimated from the training set as the 99th percentile of the absolute first-order differences of observed power. The clipped trajectory is then sequentially adjusted as

$$\tilde{P}_{t+h} = \bar{P}_{t+h-1} + \text{clip}\left(\bar{P}_{t+h} - \bar{P}_{t+h-1}, -R, R\right), \quad (24)$$

with $\tilde{P}_{t+1} = \bar{P}_{t+1}$ as initialization. This post-processing step enforces non-negativity and capacity consistency, while encouraging the final day-ahead trajectory to remain compatible with historically observed short-term variation. Since R is calibrated from historical data rather than derived from a hard physical device limit, this projection is used as a feasibility-oriented heuristic rather than a strict physical law.

A. Evaluation Metrics

To evaluate the performance of the proposed model in PV power day-ahead forecasting, comparative experiments were conducted with several benchmark algorithms. In this study, two key evaluation metrics are adopted to assess the point forecasting accuracy: Normalized Mean Absolute Error (NMAE) and Normalized Root Mean Square Error (NRMSE).

NMAE provides a robust and linear estimate of the average absolute prediction error relative to the plant's capacity, reflecting the overall baseline accuracy of the model. Conversely, NRMSE is inherently more sensitive to large forecasting deviations (e.g., unexpected power drops induced by sudden cloud cover) because the errors are squared before being averaged. By normalizing these metrics using the plant's rated power (P_r), we ensure a standardized and fair evaluation that is independent of the specific installation scale. The metrics are mathematically defined as follows:

$$\text{NMAE} = \frac{1}{H} \sum_{h=1}^H \frac{|\hat{P}_{t+h} - P_{t+h}|}{P_r} \times 100\% \quad (25)$$

$$\text{NRMSE} = \frac{1}{P_r} \sqrt{\frac{1}{H} \sum_{h=1}^H (\hat{P}_{t+h} - P_{t+h})^2} \times 100\% \quad (26)$$

where H represents the total number of time steps in the forecasting horizon, $\hat{P}_{t+h}$ is the predicted PV power at future time step $t+h$, P_{t+h} is the corresponding actual power observation (ground truth), and P_r denotes the rated active power capacity of the PV system. Lower values for both NMAE and NRMSE indicate superior forecasting performance.

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